Lecture Notes: Fourier Analysis of Discrete-Time Signals

Course: 2110203 Computer Engineering Mathematics II
Lecture 04

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Contents

1 Introduction

1.1 The Four Fourier Representations

Fourier analysis expresses a signal as a combination of complex exponentials. Which representation applies depends on two questions: is the independent variable continuous or discrete, and is the signal periodic or aperiodic?

Domain	**Signal**	**Time**	**Frequency**
Continuous-time	Periodic	$x(t)$	a_k
Continuous-time	Aperiodic	$x(t)$	$X(j\omega)$
Discrete-time	Periodic	$x[n]$	a_k
Discrete-time	Aperiodic	$x[n]$	$X(e^{j\omega})$

Table 1: Time-domain and frequency-domain representations. The first two rows were covered in the continuous-time lectures; this lecture develops the last two.

1.2 Recap: The Continuous-Time Fourier Series

For a continuous-time periodic signal $x(t)$ with fundamental period T and fundamental angular frequency $\omega_0 = 2\pi/T$, the Fourier series pair is

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t},$$

$$a_k = \frac{1}{T}\int_{\langle T\rangle} x(t)e^{-jk\omega_0 t}\,dt = \frac{1}{T}\int_0^T x(t)e^{-jk(2\pi/T)t}\,dt.$$

The summation runs over *infinitely many* harmonics k. This is the single most important point of contrast with the discrete-time case developed below, where the sum collapses to exactly N terms.

1.3 Scope of This Lecture

This lecture covers:

- the Fourier series representation of discrete-time periodic signals (the DTFS),
- the discrete-time Fourier transform (DTFT) and its derivation as a limiting case of the DTFS,
- the DTFT of periodic signals,
- the properties of the DTFT.

2 Discrete-Time Fourier Series

2.1 Periodicity of Discrete-Time Complex Exponentials

A discrete-time signal $x[n]$ is periodic with period N if

$$x[n] = x[n+N]$$

for all n, where N is a positive integer. The smallest such N is the fundamental period.

The complex exponential $e^{j(2\pi/N)n}$ is periodic with period N, since

$$e^{j(2\pi/N)(n+N)} = e^{j(2\pi/N)n}e^{j2\pi} = e^{j(2\pi/N)n}.$$

The set of all discrete-time complex exponential signals that are periodic with period N is therefore

$$\phi_k[n] = e^{j(k\omega_0)n} = e^{jk(2\pi/N)n}, \qquad k = 0, \pm1, \pm2, \ldots \tag{1}$$

where $\omega_0 = 2\pi/N$. All of these signals have fundamental frequencies that are integer multiples of $2\pi/N$, and so they are **harmonically related**.

2.2 Only N of the Harmonics Are Distinct

Discrete-time complex exponentials whose frequencies differ by an integer multiple of 2π are *identical*. Substituting $k + rN$ for k in (1), where r is any integer,

$$\phi_{k+rN}[n] = e^{j(k+rN)(2\pi/N)n} = e^{jk(2\pi/N)n} \underbrace{e^{j2\pi rn}}_{=1} = \phi_k[n],$$

because rn is an integer and $e^{j2\pi(\text{integer})} = 1$. Therefore

$$\phi_k[n] = \phi_{k+rN}[n]. \tag{2}$$

In particular $\phi_0[n] = \phi_N[n]$ and $\phi_1[n] = \phi_{N+1}[n]$.

This differs fundamentally from the continuous-time case. In continuous time the signals $e^{jk\omega_0 t}$ are all distinct for distinct k: there are infinitely many harmonics. In discrete time there are only N distinct harmonics, and the sequence $\phi_k[n]$ is itself periodic in the harmonic index k with period N.

Figure 1 illustrates this. In the left column the continuous-time exponentials oscillate faster and faster as k increases without bound. In the right column the same continuous-time signal is drawn in red and the discrete-time sequence is shown as samples taken on it at integer n. For $k = 1$ and $k = 3$ the samples follow the red curve faithfully. For $k = 5$ and $k = 7$ they do not: the samples land on a *slower* curve, drawn dashed in green. Reading only the sample values, $k = 7$ is indistinguishable from $k = 1$ and $k = 5$ is indistinguishable from $k = 3$. At $k = 8$ $(= N)$ the sequence is identical to the constant sequence at $k = 0$.

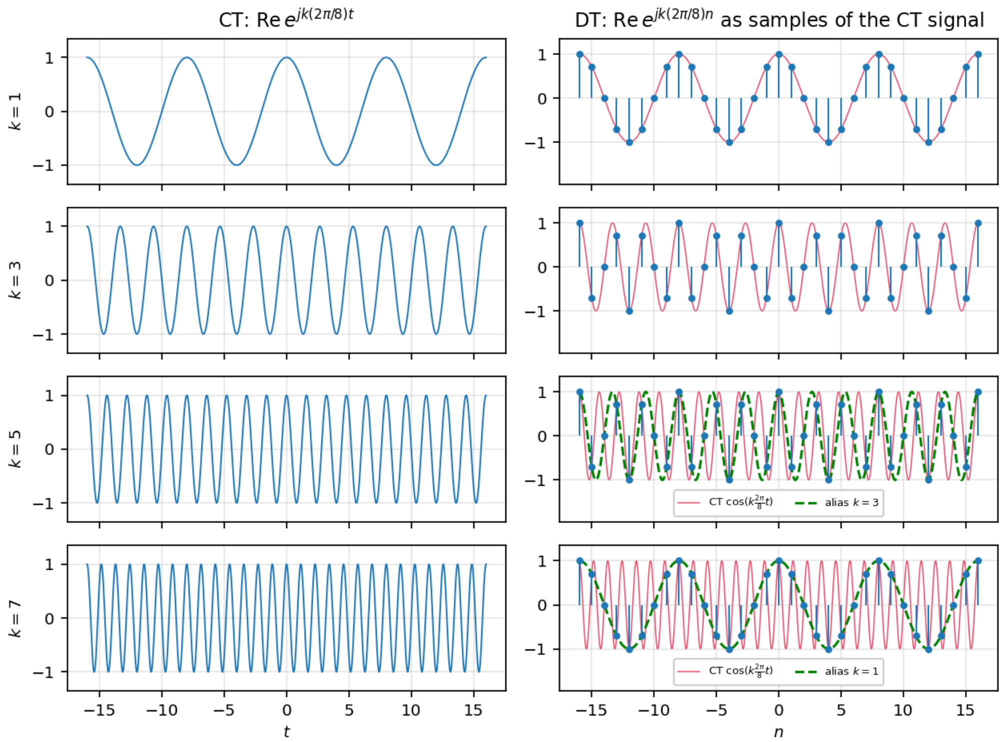


Figure 1: Real parts of the harmonics $e^{jk(2\pi/8)t}$ (continuous time, left) and $e^{jk(2\pi/8)n}$ (discrete time, right) for $k = 1, 3, 5, 7$. In the right column the red curve is the underlying continuous-time signal and the stems are its samples at integer n. The green dashed curve is the alias: the lowest-frequency harmonic passing through the same samples. For $k = 7$ the samples trace the $k = 1$ harmonic, and for $k = 5$ they trace the $k = 3$ harmonic, so these pairs are the same discrete-time signal.

2.3 The Highest Frequency in Discrete Time is π

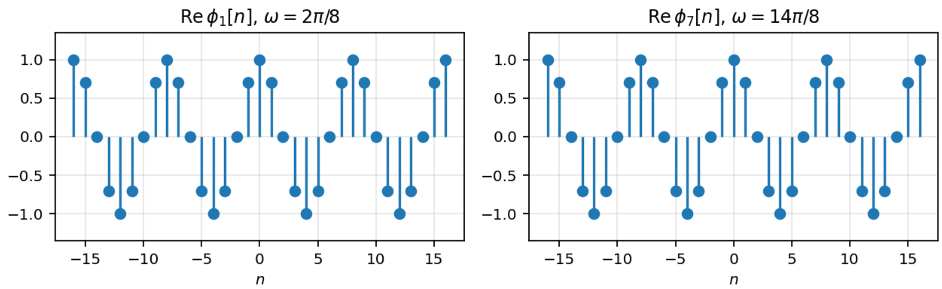
Consider $N = 8$ and compare $k = 1$ with $k = 7$. Using $\cos\theta = \cos(2\pi n - \theta)$ for integer n,

$$\cos\left(7 \cdot \frac{2\pi}{8} n\right) = \cos\left(2\pi n - \frac{2\pi}{8} n\right) = \cos\left(\frac{2\pi}{8} n\right),$$

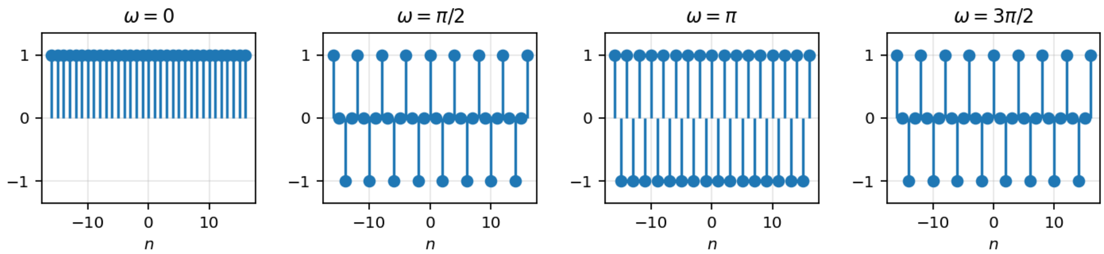
so the $k = 7$ harmonic oscillates at the *same visible rate* as the $k = 1$ harmonic, not seven times faster. The same argument applied to the sine part gives $\phi_7[n] = \phi_{-1}[n] = \phi_1^*[n]$.

The consequence is that as ω increases from 0, the rate of oscillation of $e^{j\omega n}$ increases only up to $\omega = \pi$, and then *decreases* again back to zero at $\omega = 2\pi$. So:

- **Low frequencies** in discrete time are values of ω near 0, or near any even multiple of π.
- **High frequencies** in discrete time are values of ω near π, or near any odd multiple of π.
- In continuous time the frequency can reach ∞; in discrete time the highest distinguishable frequency is π.



(a) $\phi_1[n]$ and $\phi_7[n]$ for $N = 8$: identical real parts.



(b) $\cos(\omega n)$ for increasing ω. The oscillation is fastest at $\omega = \pi$ and slows again beyond it.

Figure 2: The rate of oscillation of a discrete-time exponential is maximal at $\omega = \pi$.

2.4 Derivation of the Fourier Series Coefficients

Since only N of the $\phi_k[n]$ are distinct, a periodic sequence $x[n]$ with period N is represented by a linear combination of just N harmonics:

$$x[n] = \sum_{k=\langle N\rangle} a_k \phi_k[n] = \sum_{k=\langle N\rangle} a_k e^{jk\omega_0 n} = \sum_{k=\langle N\rangle} a_k e^{jk(2\pi/N)n}. \tag{3}$$

The notation $k = \langle N\rangle$ means that k runs over *any* N consecutive integers: $k = 0, 1, \ldots, N-1$ works, and so does $k = 3, 4, \ldots, N+2$. By (2) the result is the same. Equation (3) is the **discrete-time Fourier series**, and the a_k are the **Fourier series coefficients**.

To find the a_k, multiply both sides of (3) by $e^{-jr(2\pi/N)n}$ and sum over N consecutive values of n:

$$\sum_{n=\langle N\rangle} x[n]e^{-jr(2\pi/N)n} = \sum_{n=\langle N\rangle} \sum_{k=\langle N\rangle} a_k e^{jk(2\pi/N)n} e^{-jr(2\pi/N)n}. \tag{4}$$

Both sums are finite, so the order of summation may be interchanged:

$$\sum_{n=\langle N\rangle} x[n]e^{-jr(2\pi/N)n} = \sum_{k=\langle N\rangle} a_k \sum_{n=\langle N\rangle} e^{j(k-r)(2\pi/N)n}. \tag{5}$$

The inner sum is a geometric series in n with ratio $\alpha = e^{j(k-r)(2\pi/N)}$. Two cases arise.

Case 1: $k-r = 0, \pm N, \pm 2N, \ldots$ Then $(k-r)/N$ is an integer, so $\alpha = e^{j2\pi(\text{integer})} = 1$ and every one of the N terms equals 1. The sum is N.

Case 2: otherwise Then $\alpha \neq 1$, and summing N consecutive terms of a geometric series gives

$$\sum_{n=\langle N\rangle} \alpha^n = \alpha^{n_0}\frac{1-\alpha^N}{1-\alpha} = 0,$$

because $\alpha^N = e^{j(k-r)2\pi} = 1$ makes the numerator vanish while the denominator does not. Combining the two cases,

$$\sum_{n=\langle N\rangle} e^{j(k-r)(2\pi/N)n} = \begin{cases} N, & (k-r) = 0, \pm N, \pm 2N, \ldots \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

This is the **orthogonality relation** for discrete-time complex exponentials. Within one period of k, exactly one term of the outer sum in (5) survives, namely $k = r$, and it contributes Na_r. Hence

$$a_r = \frac{1}{N}\sum_{n=\langle N\rangle} x[n]e^{-jr(2\pi/N)n}. \tag{7}$$

2.5 Summary: The Discrete-Time Fourier Series Pair

$$x[n] = \sum_{k=\langle N\rangle} a_k e^{jk\omega_0 n} = \sum_{k=\langle N\rangle} a_k e^{jk(2\pi/N)n} \qquad \text{(synthesis)} \tag{8}$$

$$a_k = \frac{1}{N}\sum_{n=\langle N\rangle} x[n]e^{-jk\omega_0 n} = \frac{1}{N}\sum_{n=\langle N\rangle} x[n]e^{-jk(2\pi/N)n} \qquad \text{(analysis)} \tag{9}$$

Two structural remarks follow directly:

- Both equations are *finite* sums. Unlike the continuous-time Fourier series, there are no convergence conditions to check and no Gibbs phenomenon: the representation is exact for every periodic sequence.
- The coefficients inherit the periodicity of the harmonics: $a_k = a_{k+N}$. The frequency-domain description of a discrete-time periodic signal is itself a periodic sequence.

2.6 Properties of the Discrete-Time Fourier Series

The properties below are collected from Table 3.2 of Oppenheim, Willsky and Nawab, reproduced on the final reference slide of the lecture. Only the conjugate-symmetry result is used explicitly in the lecture body, in Example 2 of Section 2.8.

Let $x[n]$ and $y[n]$ be periodic with period N and fundamental frequency $\omega_0 = 2\pi/N$, with coefficients a_k and b_k (both periodic in k with period N).

Conjugate symmetry for real signals. If $x[n]$ is real, then

$$a_k = a_{-k}^*, \qquad \mathrm{Re}\{a_k\} = \mathrm{Re}\{a_{-k}\}, \qquad \mathrm{Im}\{a_k\} = -\mathrm{Im}\{a_{-k}\},$$
$$|a_k| = |a_{-k}|, \qquad \sphericalangle a_k = -\sphericalangle a_{-k}.$$

If $x[n]$ is real and even, a_k is real and even. If $x[n]$ is real and odd, a_k is purely imaginary and odd.

Property	Periodic signal	Fourier series coefficients
Linearity	$Ax[n] + By[n]$	$Aa_k + Bb_k$
Time shifting	$x[n - n_0]$	$a_k e^{-jk(2\pi/N)n_0}$
Frequency shifting	$e^{jM(2\pi/N)n} x[n]$	a_{k-M}
Conjugation	$x^*[n]$	a^*_{-k}
Time reversal	$x[-n]$	a_{-k}
Time scaling	$x_{(m)}[n] = \begin{cases} x[n/m], & n \text{ multiple of } m \\ 0, & \text{otherwise} \end{cases}$	$\frac{1}{m} a_k$ (period mN)
Periodic convolution	$\sum_{r=\langle N\rangle} x[r]y[n-r]$	$N a_k b_k$
Multiplication	$x[n]y[n]$	$\sum_{l=\langle N\rangle} a_l b_{k-l}$
First difference	$x[n] - x[n-1]$	$\left(1 - e^{-jk(2\pi/N)}\right) a_k$
Running sum	$\sum_{k=-\infty}^{n} x[k]$ (if $a_0 = 0$)	$\left(\dfrac{1}{1 - e^{-jk(2\pi/N)}}\right) a_k$
Even part (real $x[n]$)	$x_e[n] = \mathcal{E}v\{x[n]\}$	$\mathrm{Re}\{a_k\}$
Odd part (real $x[n]$)	$x_o[n] = \mathcal{O}d\{x[n]\}$	$j\,\mathrm{Im}\{a_k\}$

Table 2: Properties of the discrete-time Fourier series (Oppenheim, Willsky & Nawab, Table 3.2). The time-scaled signal $x_{(m)}[n]$ is periodic with period mN, and its coefficients are viewed as periodic with that same period. The running sum is finite-valued and periodic only when $a_0 = 0$.

Parseval's relation for periodic signals. The average power can be computed in either domain:

$$\frac{1}{N} \sum_{n=\langle N\rangle} |x[n]|^2 = \sum_{k=\langle N\rangle} |a_k|^2.$$

2.7 Worked Example 1: The Periodic Square Wave

Problem. Let $x[n]$ be periodic with period N, defined over one period by

$$x[n] = \begin{cases} 1, & -N_1 \le n \le N_1, \\ 0, & N_1 < |n| \le N/2. \end{cases}$$

Find the Fourier series coefficients a_k, and plot Na_k for $2N_1 + 1 = 5$ with $N = 10, 20, 40$.

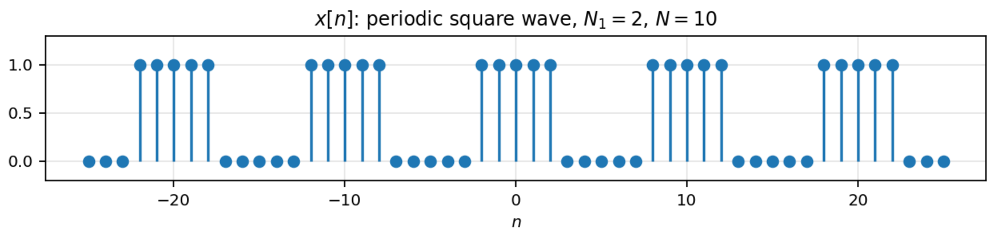


Figure 3: The periodic square wave with $N_1 = 2$ (so $2N_1 + 1 = 5$ nonzero samples per period) and $N = 10$.

Step 1: write down the analysis equation over a convenient period. Choose the interval $-N_1 \le n \le N - N_1 - 1$, which is N consecutive samples and puts the nonzero block at the start. Since $x[n] = 0$ outside $|n| \le N_1$,

$$a_k = \frac{1}{N} \sum_{n=\langle N\rangle} x[n] e^{-jk(2\pi/N)n} = \frac{1}{N} \sum_{n=-N_1}^{N_1} e^{-jk(2\pi/N)n}.$$

Step 2: shift the index so the sum starts at zero. Let $m = n + N_1$, so $n = m - N_1$, and as n runs from $-N_1$ to N_1, m runs from 0 to $2N_1$:

$$a_k = \frac{1}{N} \sum_{m=0}^{2N_1} e^{-jk(2\pi/N)(m-N_1)} = \frac{1}{N} e^{jk(2\pi/N)N_1} \sum_{m=0}^{2N_1} e^{-jk(2\pi/N)m}.$$

The factor $e^{jk(2\pi/N)N_1}$ does not depend on m and has been pulled out.

Step 3: sum the geometric series. The remaining sum has ratio $r = e^{-jk(2\pi/N)}$ and $2N_1 + 1$ terms. Provided $r \neq 1$, that is $k \neq 0, \pm N, \pm 2N, \ldots$,

$$\sum_{m=0}^{2N_1} r^m = \frac{1 - r^{\,2N_1+1}}{1 - r} = \frac{1 - e^{-jk2\pi(2N_1+1)/N}}{1 - e^{-jk(2\pi/N)}},$$

so

$$a_k = \frac{1}{N} e^{jk(2\pi/N)N_1} \left(\frac{1 - e^{-jk2\pi(2N_1+1)/N}}{1 - e^{-jk(2\pi/N)}} \right). \tag{10}$$

Step 4: symmetrise numerator and denominator. The expression is real up to a phase factor; extract half of each exponent to expose a sine. For the numerator, factor out $e^{-jk\pi(2N_1+1)/N}$:

$$1 - e^{-jk2\pi(2N_1+1)/N} = e^{-jk\pi(2N_1+1)/N} \left[e^{jk\pi(2N_1+1)/N} - e^{-jk\pi(2N_1+1)/N} \right].$$

Combining the leading phase with this factor,

$$e^{jk(2\pi/N)N_1} \, e^{-jk\pi(2N_1+1)/N} = e^{jk\pi(2N_1-2N_1-1)/N} = e^{-jk\pi/N}.$$

For the denominator, factor out $e^{-jk\pi/N}$ in the same way:

$$1 - e^{-jk(2\pi/N)} = e^{-jk\pi/N} \left[e^{jk\pi/N} - e^{-jk\pi/N} \right].$$

Step 5: apply Euler's identity and cancel. Using $e^{j\theta} - e^{-j\theta} = 2j \sin\theta$, the bracketed terms become $2j \sin\big(\pi k(2N_1+1)/N\big)$ and $2j \sin(\pi k/N)$. The common factors $e^{-jk\pi/N}$ and $2j$ cancel between numerator and denominator, leaving

$$a_k = \frac{1}{N} \frac{\sin\big[2\pi k(N_1 + \frac{1}{2})/N\big]}{\sin(\pi k/N)}, \qquad k \neq 0, \pm N, \pm 2N, \ldots \tag{11}$$

Step 6: handle the excluded values of k. When $k = 0, \pm N, \pm 2N, \ldots$, the ratio r equals 1 and Step 3 does not apply. Going back to Step 2, every term of the sum is 1, and there are $2N_1 + 1$ of them:

$$a_k = \frac{2N_1 + 1}{N}, \qquad k = 0, \pm N, \pm 2N, \ldots \tag{12}$$

This is exactly the limit of (11) as $k \to 0$, so the coefficient sequence has no actual discontinuity.

Step 7: interpret the result. Multiplying by N removes the $1/N$ scaling and lets different periods be compared on the same axes:

$$N a_k = \frac{\sin\big[2\pi k(N_1 + \frac{1}{2})/N\big]}{\sin(\pi k/N)}.$$

Figure 4 plots this for $2N_1 + 1 = 5$ and increasing N. Note:

- a_k is real (the signal is real and even), and periodic in k with period N, as predicted in Section 2.5.
- The peak value is $N a_0 = 2N_1 + 1 = 5$, the number of nonzero samples per period.

- As N increases with N_1 fixed, the *envelope* does not change shape; the samples just get packed more densely along it. Writing $\omega = 2\pi k/N$, the envelope is $\sin\big[\omega(N_1+\frac{1}{2})\big]/\sin(\omega/2)$.

That last observation is the entire idea behind Section 3: letting $N \to \infty$ turns the discrete coefficient sequence into a continuous function of ω.

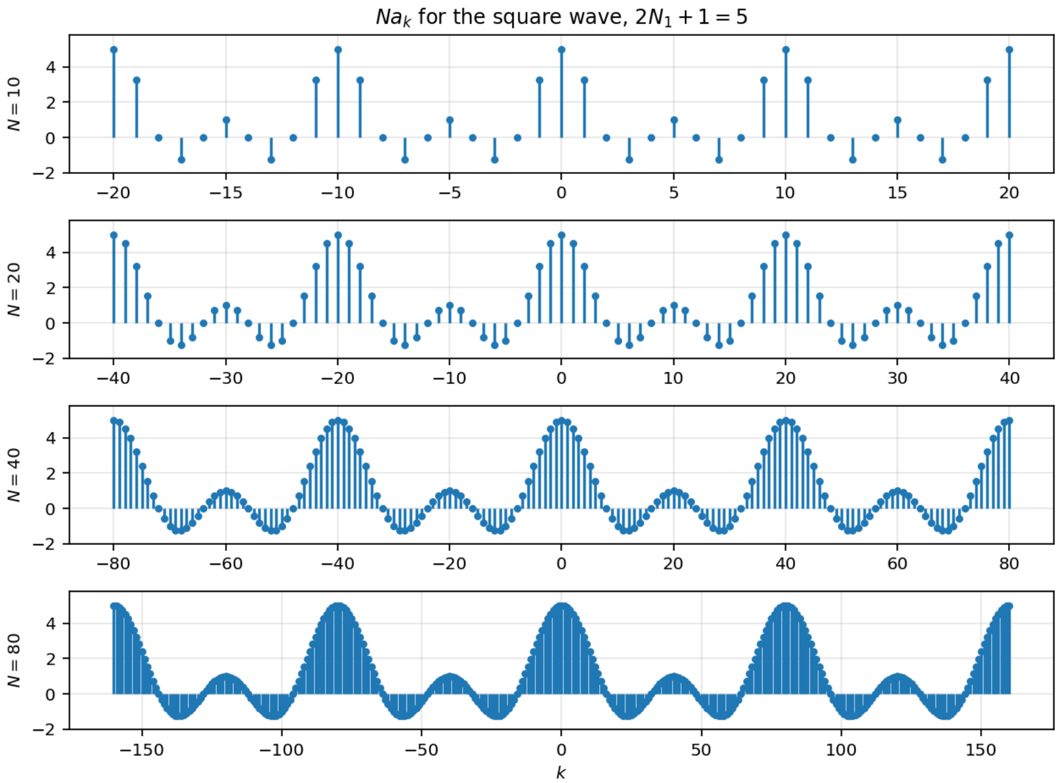


Figure 4: Scaled Fourier coefficients Na_k for the periodic square wave with $2N_1+1=5$ and $N = 10, 20, 40, 80$. The envelope is fixed; only the sample spacing $2\pi/N$ changes.

Reconstruction from partial sums. Because the discrete-time Fourier series is a finite sum, including all N terms reproduces $x[n]$ *exactly*. Figure 5 shows the partial sums $\sum_{|k|\le M} a_k e^{jk(2\pi/N)n}$ for $N = 10$: at $M = 5$ all ten distinct harmonics are present and the square wave is recovered with no overshoot at the edges.

2.8 Worked Example 2: A Sum of Sinusoids

Problem. Find the Fourier series coefficients of

$$x[n] = 1 + \sin\left(\frac{2\pi}{N}n\right) + 3\cos\left(\frac{2\pi}{N}n\right) + \cos\left(\frac{4\pi}{N}n + \frac{\pi}{2}\right).$$

Step 1: identify the fundamental frequency. The three sinusoidal terms have frequencies $2\pi/N$, $2\pi/N$ and $4\pi/N = 2(2\pi/N)$. All are integer multiples of $\omega_0 = 2\pi/N$, so $x[n]$ is periodic with period N and the harmonics present are $k = 0, \pm1, \pm2$.

Step 2: expand each term with Euler's formula. Rather than evaluating the analysis equation (9), it is faster to write $x[n]$ directly in the form of the synthesis equation (8) and read the coefficients off. Using $\sin\theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$ and $\cos\theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$:

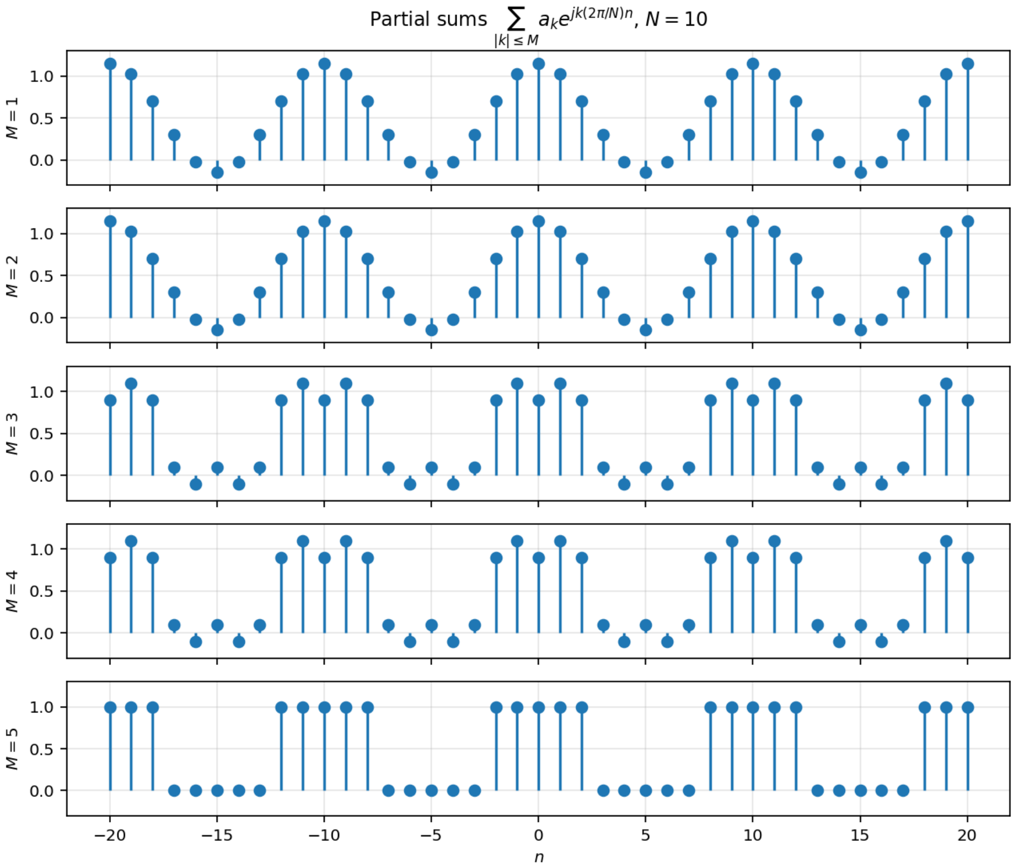


Figure 5: Partial sums of the discrete-time Fourier series for the square wave, $N = 10$, $N_1 = 2$. Reconstruction is exact once all N harmonics are included.

$$x[n] = 1 + \frac{1}{2j}\left[e^{j(2\pi/N)n} - e^{-j(2\pi/N)n}\right] + \frac{3}{2}\left[e^{j(2\pi/N)n} + e^{-j(2\pi/N)n}\right] + \frac{1}{2}\left[e^{j(4\pi n/N+\pi/2)} + e^{-j(4\pi n/N+\pi/2)}\right].$$

Step 3: collect terms by harmonic index. Separate the constant phase in the last term,

$$e^{\pm j(4\pi n/N+\pi/2)} = e^{\pm j\pi/2}\, e^{\pm j2(2\pi/N)n},$$

and group by harmonic:

$$x[n] = 1 + \left(\frac{3}{2} + \frac{1}{2j}\right) e^{j(2\pi/N)n} + \left(\frac{3}{2} - \frac{1}{2j}\right) e^{-j(2\pi/N)n} + \frac{1}{2}e^{j\pi/2}\, e^{j2(2\pi/N)n} + \frac{1}{2}e^{-j\pi/2}\, e^{-j2(2\pi/N)n}.$$

Step 4: read off the coefficients. Comparing with $x[n] = \sum_k a_k e^{jk(2\pi/N)n}$, and using $1/j = -j$ together with $e^{\pm j\pi/2} = \pm j$:

$$a_0 = 1, \qquad a_1 = \frac{3}{2} + \frac{1}{2j} = \frac{3}{2} - \frac{1}{2}j, \qquad a_{-1} = \frac{3}{2} - \frac{1}{2j} = \frac{3}{2} + \frac{1}{2}j,$$
$$a_2 = \frac{1}{2}j, \qquad a_{-2} = -\frac{1}{2}j,$$

and $a_k = 0$ for all other k within one period. Outside that period the coefficients repeat with period N.

Step 5: check the symmetry. The signal $x[n]$ is real, so the conjugate symmetry property of Table 2 requires $a_{-k} = a_k^*$. Checking: $a_{-1} = \frac{3}{2} + \frac{1}{2}j = (\frac{3}{2} - \frac{1}{2}j)^* = a_1^*$, and $a_{-2} = -\frac{1}{2}j = (\frac{1}{2}j)^* = a_2^*$. The check passes.

Step 6: magnitude and phase. For $N = 10$,

$$|a_1| = \sqrt{\left(\tfrac{3}{2}\right)^2 + \left(\tfrac{1}{2}\right)^2} = \frac{\sqrt{10}}{2}, \qquad \sphericalangle a_1 = -\arctan\left(\frac{1/2}{3/2}\right) = -\arctan\frac{1}{3},$$

$$|a_2| = \frac{1}{2}, \qquad \sphericalangle a_2 = \frac{\pi}{2}.$$

Figure 6 shows the four standard views of the coefficient sequence. As predicted, the real part and the magnitude are even in k, while the imaginary part and the phase are odd, and the whole pattern repeats every $N = 10$.

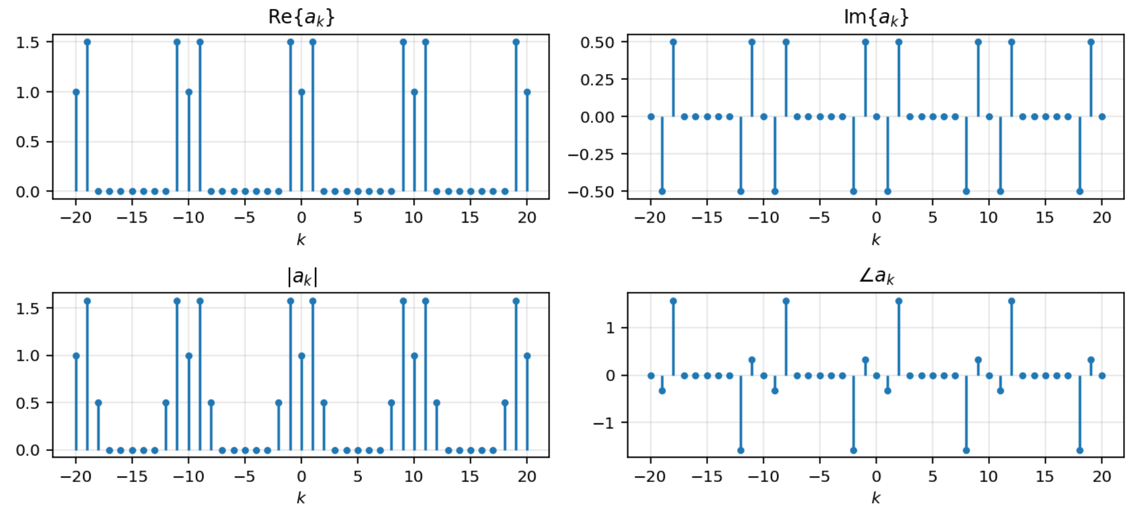


Figure 6: Fourier series coefficients of Example 2 for $N = 10$: real part, imaginary part, magnitude and phase. Even symmetry of $\mathrm{Re}\{a_k\}$ and $|a_k|$, odd symmetry of $\mathrm{Im}\{a_k\}$ and $\sphericalangle a_k$, and periodicity in k with period N.

2.9 Comparison: Continuous-Time and Discrete-Time Fourier Series

	Continuous time	**Discrete time**
Synthesis	$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}$	$x[n] = \sum_{k=\langle N\rangle} a_k e^{jk(2\pi/N)n}$
Analysis	$a_k = \frac{1}{T}\int_{\langle T\rangle} x(t) e^{-jk(2\pi/T)t} dt$	$a_k = \frac{1}{N}\sum_{n=\langle N\rangle} x[n] e^{-jk(2\pi/N)n}$
Number of harmonics	infinite	exactly N
Coefficients	aperiodic in k	periodic in k with period N
Convergence	conditions apply; Gibbs phenomenon	finite sum; always exact
Highest frequency	unbounded	$\omega = \pi$

Table 3: Fourier series in continuous and discrete time.

3 The Discrete-Time Fourier Transform

The Fourier series of Section 2 applies only to periodic sequences. To represent an *aperiodic* sequence we use the same device as in continuous time: build a periodic signal whose period contains the aperiodic signal, take its Fourier series, then let the period grow without bound.

3.1 From the Fourier Series to the Transform

Step 1: construct a periodic extension. Let $x[n]$ have finite duration, meaning $x[n] = 0$ outside some range $-N_1 \le n \le N_2$. Construct a periodic sequence $\tilde{x}[n]$ with period N that equals $x[n]$ over one period, as in Figure 7. Provided $N > N_1 + N_2$, no overlap occurs, and as N grows, $\tilde{x}[n] = x[n]$ over a longer and longer interval. In the limit $N \to \infty$ we have $\tilde{x}[n] = x[n]$ for any finite n.

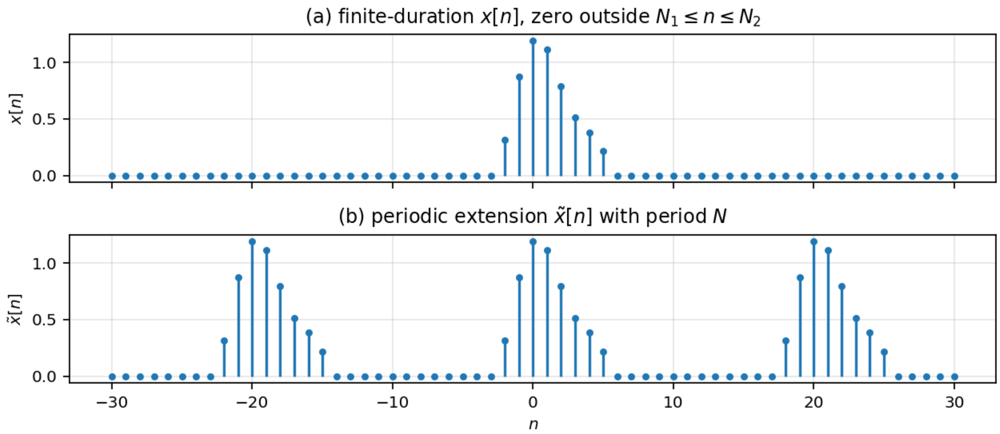


Figure 7: (a) A finite-duration signal $x[n]$. (b) The periodic signal $\tilde{x}[n]$ constructed to equal $x[n]$ over one period. Letting $N \to \infty$ recovers $x[n]$.

Step 2: write the Fourier series of the extension. Since $\tilde{x}[n]$ is periodic with period N, Section 2.5 gives

$$\tilde{x}[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}, \qquad a_k = \frac{1}{N} \sum_{n=\langle N \rangle} \tilde{x}[n] e^{-jk(2\pi/N)n}.$$

Step 3: replace $\tilde{x}[n]$ by $x[n]$ and extend the limits. Choose the summation interval for a_k to be the one period containing the nonzero block. Over that interval $\tilde{x}[n] = x[n]$, so

$$a_k = \frac{1}{N} \sum_{n=-N_1}^{N_2} x[n] e^{-jk(2\pi/N)n} = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk(2\pi/N)n}.$$

Extending the limits to $\pm\infty$ costs nothing, because $x[n]$ vanishes outside $-N_1 \le n \le N_2$. This is the crucial move: the sum no longer refers to N at all except through the exponent.

Step 4: define the envelope function. Define a function of the continuous variable ω,

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}. \tag{13}$$

Comparing with Step 3, the Fourier coefficients are *samples* of this function:

$$a_k = \frac{1}{N} X(e^{jk\omega_0}), \qquad \omega_0 = \frac{2\pi}{N}, \tag{14}$$

where ω_0 is the spacing between adjacent samples in the frequency domain. This is precisely the behaviour observed in Example 1 of Section 2.7: the envelope stayed fixed while increasing N packed the samples more densely along it.

Step 5: substitute back into the synthesis equation. Using (14),

$$\tilde{x}[n] = \frac{1}{N} \sum_{k=\langle N \rangle} X(e^{jk\omega_0}) e^{jk\omega_0 n}.$$

Since $N = 2\pi/\omega_0$, we may replace $1/N$ by $\omega_0/2\pi$:

$$\tilde{x}[n] = \frac{1}{2\pi} \sum_{k=\langle N \rangle} X(e^{jk\omega_0}) e^{jk\omega_0 n}\, \omega_0. \tag{15}$$

Step 6: take the limit. Equation (15) is a Riemann sum. Each term is the value of the function $X(e^{j\omega})e^{j\omega n}$ evaluated at $\omega = k\omega_0$, multiplied by the width ω_0 of the interval it represents. As $N \to \infty$ we have $\omega_0 \to 0$, the sum passes to an integral, and $\tilde{x}[n] \to x[n]$.

The summation runs over N consecutive values of k, each contributing an interval of width $\omega_0 = 2\pi/N$, so the total interval of integration always has width $N \cdot (2\pi/N) = 2\pi$, no matter how large N becomes. Hence

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n}\, d\omega. \tag{16}$$

Step 7: note that the interval may be any 2π window. From (13), $X(e^{j\omega})$ is periodic in ω with period 2π (Section 3.4), and so is $X(e^{j\omega})e^{j\omega n}$, since $e^{j2\pi n} = 1$ for integer n. The integral in (16) may therefore be taken over *any* interval of length 2π; $[0, 2\pi]$ and $[-\pi, \pi]$ are the usual choices.

3.2 The Discrete-Time Fourier Transform Pair

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n}\, d\omega \qquad \text{(synthesis)} \tag{17}$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \qquad \text{(analysis)} \tag{18}$$

Equation (17) is the **synthesis** equation and (18) the **analysis** equation. The function $X(e^{j\omega})$ is called the **spectrum** of $x[n]$, since it describes how $x[n]$ is composed from complex exponentials at different frequencies. The pair is written

$$x[n] \stackrel{\mathcal{F}}{\longleftrightarrow} X(e^{j\omega}).$$

Note the asymmetry with the Fourier series: the time-domain signal is a sequence, but the frequency-domain description is a *continuous* function of ω. Both the DTFS and the DTFT of an aperiodic signal are periodic in the frequency variable, but only the DTFS has a discrete frequency variable.

3.3 Convergence Issues

Because (18) is an infinite sum, it need not converge for every sequence. It does converge if $x[n]$ is **absolutely summable**,

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty,$$

or if the sequence has **finite energy**,

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty.$$

The first condition guarantees uniform convergence; the second guarantees convergence in the mean-square sense.

By contrast, the synthesis equation (17) integrates over a *finite* interval of length 2π, so there are no convergence issues associated with it. This is the reverse of the continuous-time situation, where the synthesis integral runs over all ω and the analysis integral over a finite period.

3.4 Periodicity, and Where Low and High Frequencies Live

Replacing ω by $\omega + 2\pi$ in (18),

$$X(e^{j(\omega+2\pi)}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \underbrace{e^{-j2\pi n}}_{=1} = X(e^{j\omega}),$$

so the DTFT is always periodic in ω with period 2π. This is not an accident of a particular signal: it is inherited from $\phi_k[n] = \phi_{k+rN}[n]$ in Section 2.2, and it means that all the information in $X(e^{j\omega})$ is contained in any single 2π interval.

Consequently, $\omega = 0$ and $\omega = 2\pi$ describe the same signal. Following Section 2.3:

- **Low frequencies** are values of ω near 0, or near any even multiple of π.
- **High frequencies** are values of ω near π, or near any odd multiple of π.

This is why a discrete-time lowpass filter has passbands centred at even multiples of π and a highpass filter has passbands centred at odd multiples of π, a fact used in Section 5.3.

3.5 Worked Example 3: The Decaying Exponential $a^n u[n]$

Problem. Find the DTFT of $x[n] = a^n u[n]$ with $|a| < 1$.

Step 1: apply the analysis equation. Since $u[n] = 0$ for $n < 0$, the lower limit becomes $n = 0$:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} a^n u[n]e^{-j\omega n} = \sum_{n=0}^{\infty} a^n e^{-j\omega n}.$$

Step 2: recognise a geometric series. Group the two exponentials into a single power of n:

$$X(e^{j\omega}) = \sum_{n=0}^{\infty} \left(ae^{-j\omega}\right)^n.$$

Step 3: check the convergence condition and sum. The infinite geometric series $\sum_{n=0}^{\infty} r^n = 1/(1-r)$ requires $|r| < 1$. Here $r = ae^{-j\omega}$, and since $|e^{-j\omega}| = 1$ for real ω, we have $|r| = |a| < 1$ by assumption. The series converges for every ω, and

$$X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}, \qquad |a| < 1. \tag{19}$$

The condition $|a| < 1$ is exactly what makes $x[n]$ absolutely summable, so the convergence requirement of Section 3.3 is satisfied.

Step 4: extract magnitude and phase. Write the denominator in rectangular form using Euler's formula:

$$1 - ae^{-j\omega} = 1 - a\cos\omega + ja\sin\omega.$$

Taking the modulus of the reciprocal,

$$\left|X(e^{j\omega})\right| = \frac{1}{\sqrt{(1-a\cos\omega)^2 + (a\sin\omega)^2}} = \frac{1}{\sqrt{1 + a^2 - 2a\cos\omega}},$$

where the last step uses $\cos^2\omega + \sin^2\omega = 1$. The phase is minus the argument of the denominator:

$$\sphericalangle X(e^{j\omega}) = -\arctan\left(\frac{a\sin\omega}{1 - a\cos\omega}\right).$$

Step 5: interpret. Figure 8 plots the results for three values of a.

- For $a > 0$ the magnitude peaks at $\omega = 0$ with value $1/(1-a)$ and is smallest at $\omega = \pi$ with value $1/(1+a)$. The signal decays smoothly, so its energy sits at low frequencies: this is a **lowpass** characteristic.
- For $a < 0$ the signal alternates in sign from sample to sample, and the magnitude peaks at $\omega = \pi$ instead: a **highpass** characteristic. This is consistent with Section 3.4, where π was identified as the highest discrete-time frequency.
- As $|a| \to 1$ the decay becomes slower, the signal spreads out in time, and the peak of $|X(e^{j\omega})|$ becomes taller and narrower. Compare $a = 0.5$ with $a = 0.86$ in the figure.
- Every plot repeats with period 2π, as required by Section 3.4.

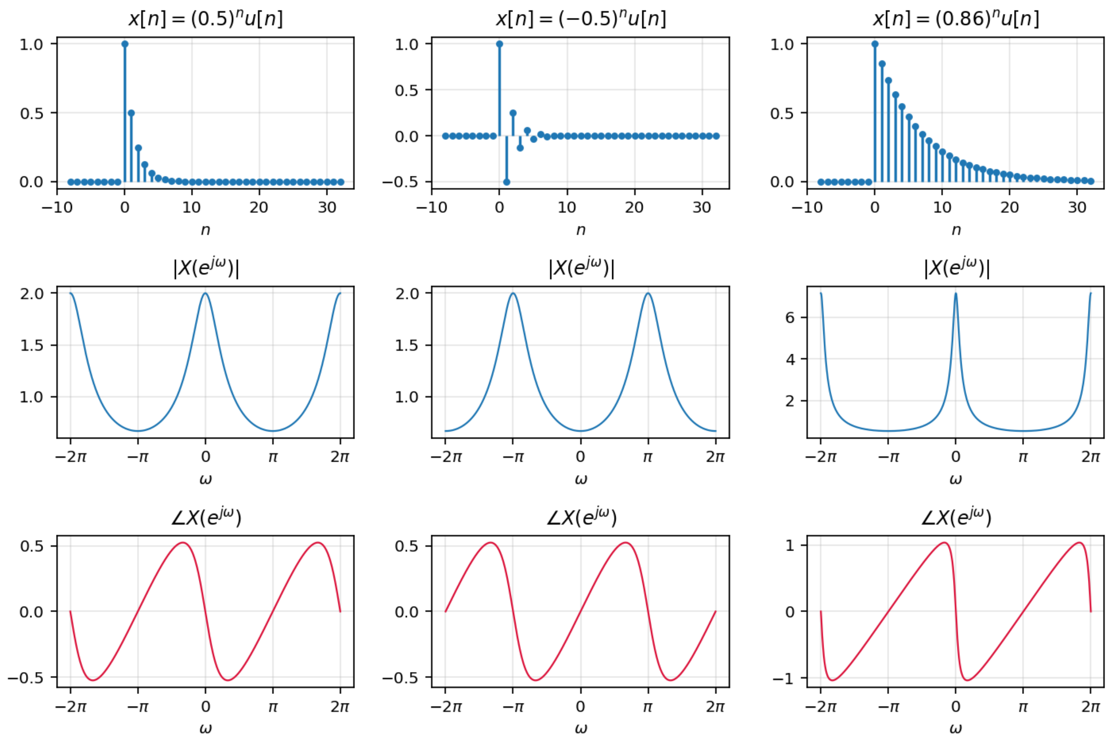


Figure 8: The signal $a^n u[n]$ (top) with the magnitude (middle) and phase (bottom) of its DTFT, for $a = 0.5$, $a = -0.5$ and $a = 0.86$. Positive a gives a lowpass spectrum, negative a a highpass spectrum, and larger $|a|$ a narrower peak.

3.6 Worked Example 4: The Rectangular Pulse

Problem. Find the DTFT of the rectangular pulse

$$x[n] = \begin{cases} 1, & |n| \le N_1, \\ 0, & |n| > N_1. \end{cases}$$

Step 1: apply the analysis equation. Only $2N_1 + 1$ terms are nonzero:

$$X(e^{j\omega}) = \sum_{n=-N_1}^{N_1} e^{-j\omega n}.$$

Step 2: shift the index. Let $m = n + N_1$, so $n = m - N_1$ and m runs from 0 to $2N_1$:

$$X(e^{j\omega}) = \sum_{m=0}^{2N_1} e^{-j\omega(m-N_1)} = e^{j\omega N_1} \sum_{m=0}^{2N_1} e^{-j\omega m}.$$

Step 3: sum the finite geometric series. With ratio $r = e^{-j\omega}$ and $2N_1 + 1$ terms, and assuming $r \neq 1$,

$$X(e^{j\omega}) = e^{j\omega N_1} \left(\frac{1 - e^{-j\omega(2N_1+1)}}{1 - e^{-j\omega}} \right).$$

Step 4: symmetrise and cancel. This is the same manipulation as Steps 4–5 of Example 1 in Section 2.7. Factor $e^{-j\omega(2N_1+1)/2}$ from the numerator bracket and $e^{-j\omega/2}$ from the denominator bracket. The leading phase combines as

$$e^{j\omega N_1} e^{-j\omega(2N_1+1)/2} = e^{-j\omega/2},$$

which cancels against the $e^{-j\omega/2}$ pulled from the denominator. Applying $e^{j\theta} - e^{-j\theta} = 2j \sin\theta$ to both brackets and cancelling the common $2j$:

$$\boxed{X(e^{j\omega}) = \frac{\sin\left[\omega\left(N_1 + \frac{1}{2}\right)\right]}{\sin(\omega/2)}.} \tag{20}$$

Step 5: check the value at $\omega = 0$. At $\omega = 0$ the excluded case $r = 1$ applies, and Step 2 gives a sum of $2N_1 + 1$ ones. Taking the limit of (20) instead, both sine arguments are small, so $\sin\theta \approx \theta$ and

$$\lim_{\omega \to 0} \frac{\omega(N_1 + \frac{1}{2})}{\omega/2} = 2N_1 + 1,$$

which agrees. Figure 9 shows the transform for $N_1 = 2$, with peak value 5.

Step 6: compare with the Fourier series result. Equation (20) is exactly the envelope identified in Section 2.7: setting $\omega = 2\pi k/N$ in (20) reproduces Na_k for the periodic square wave. The DTFT of one pulse is the envelope; the Fourier series coefficients of the periodic train of pulses are samples of that envelope, spaced $2\pi/N$ apart. This confirms the mechanism derived in Section 3.1.

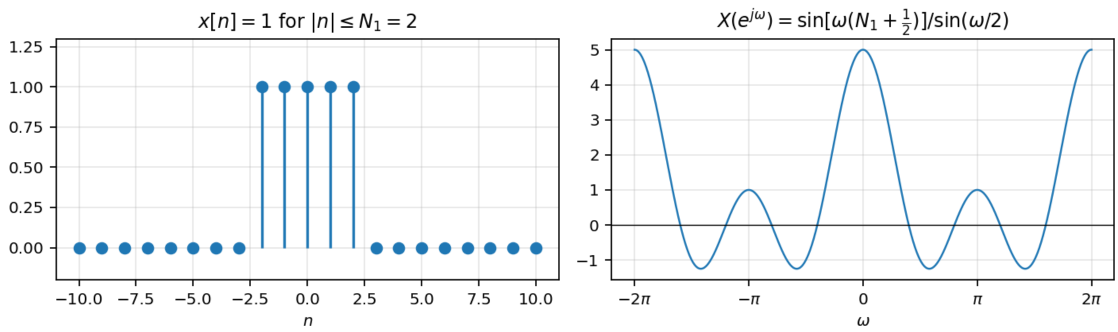


Figure 9: The rectangular pulse with $N_1 = 2$ and its DTFT. The transform is real (the signal is real and even), periodic with period 2π, and equals $2N_1 + 1 = 5$ at $\omega = 0$.

3.7 Worked Example 5: The Unit Impulse and the Ideal Lowpass Filter

Part I: the unit impulse. Let $x[n] = \delta[n]$. Applying the analysis equation and using the sampling property of the discrete impulse, which selects the $n = 0$ term of the sum,

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \delta[n]e^{-j\omega n} = e^{-j\omega \cdot 0} = 1.$$

The unit impulse has a Fourier transform consisting of equal contributions at all frequencies:

$$\delta[n] \xleftrightarrow{\mathcal{F}} 1. \tag{21}$$

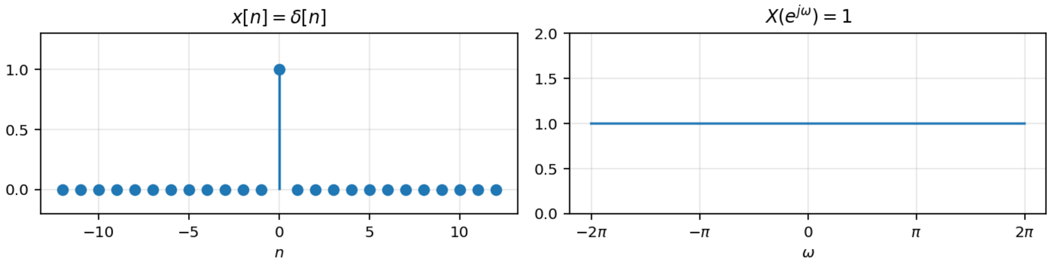


Figure 10: The unit impulse and its DTFT: a constant across all ω.

Part II: the ideal lowpass filter. Now go the other way and start in the frequency domain. Let $X(e^{j\omega})$ be 1 for $|\omega| \le W$ and 0 for $W < |\omega| \le \pi$, extended periodically with period 2π as required by Section 3.4. Find $x[n]$.

Step 1: apply the synthesis equation over one period. Choose the interval $[-\pi, \pi]$. Since $X(e^{j\omega})$ vanishes outside $|\omega| \le W$,

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}\,d\omega = \frac{1}{2\pi}\int_{-W}^{W} e^{j\omega n}\,d\omega.$$

Step 2: integrate. For $n \ne 0$, the antiderivative of $e^{j\omega n}$ with respect to ω is $e^{j\omega n}/(jn)$:

$$x[n] = \frac{1}{2\pi}\left.\frac{e^{j\omega n}}{jn}\right|_{\omega=-W}^{\omega=W} = \frac{1}{2\pi}\left(\frac{e^{jWn}}{jn} - \frac{e^{-jWn}}{jn}\right).$$

Step 3: apply Euler's identity. Using $e^{j\theta} - e^{-j\theta} = 2j\sin\theta$ with $\theta = Wn$,

$$x[n] = \frac{1}{2\pi}\cdot\frac{2j\sin(Wn)}{jn} = \frac{\sin(Wn)}{\pi n}.$$

Step 4: handle $n = 0$. At $n = 0$ the integrand is 1, so $x[0] = \frac{1}{2\pi}\int_{-W}^{W} d\omega = W/\pi$. This is the limit of $\sin(Wn)/(\pi n)$ as $n \to 0$, so the formula extends continuously. Thus

$$\frac{\sin(Wn)}{\pi n} \xleftrightarrow{\mathcal{F}} X(e^{j\omega}) = \begin{cases} 1, & 0 \le |\omega| \le W, \\ 0, & W < |\omega| \le \pi, \end{cases} \qquad 0 < W < \pi, \tag{22}$$

with $X(e^{j\omega})$ periodic in ω with period 2π.

Step 5: interpret. Figure 11 shows $x[n]$ for four values of W. A narrow band of frequencies ($W = \pi/4$) corresponds to a sequence that decays slowly and spreads widely in n; a wide band ($W = \pi$) concentrates the sequence at the origin. At $W = \pi$ the passband covers the whole period, $X(e^{j\omega}) = 1$ everywhere, and $x[n] = \delta[n]$, recovering Part I.

This inverse relationship between width in time and width in frequency is the same trade-off seen in Example 3, where slower decay produced a narrower spectral peak. Note also the pairing with Example 4: a rectangle in *time* transforms to a Dirichlet-type ratio of sines in frequency, while a rectangle in *frequency* transforms to a sampled sinc in time.

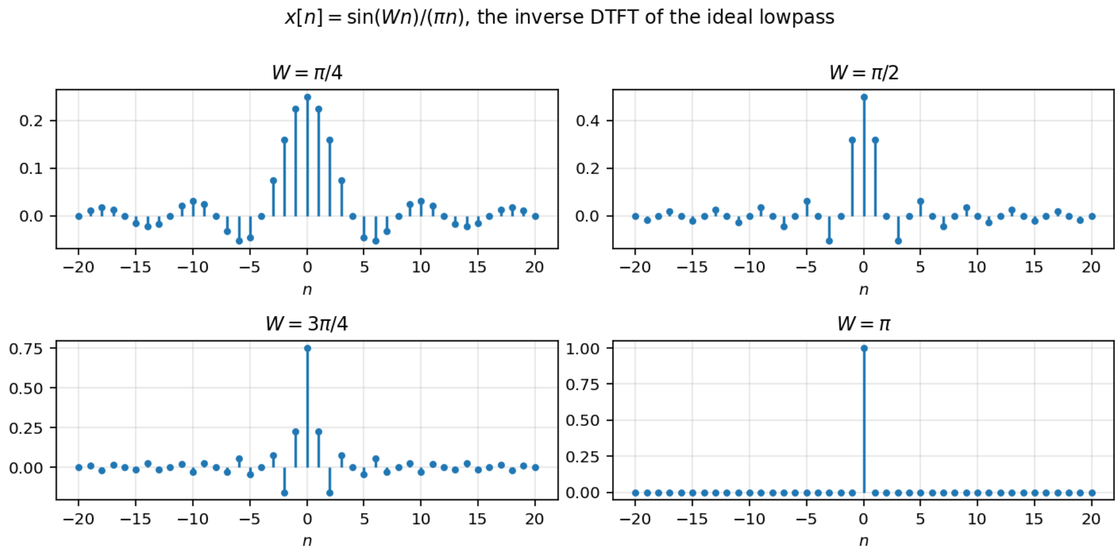


Figure 11: The inverse DTFT of the ideal lowpass characteristic, $x[n] = \sin(Wn)/(\pi n)$, for $W = \pi/4, \pi/2, 3\pi/4, \pi$. Widening the passband concentrates the sequence; at $W = \pi$ it collapses to $\delta[n]$.

4 The Fourier Transform of Periodic Signals

A periodic sequence is not absolutely summable and does not have finite energy, so the convergence conditions of Section 3.3 fail. As in continuous time, the difficulty is resolved by admitting impulses in the frequency domain. Periodic signals can then be handled inside the DTFT framework, and the Fourier series of Section 2 becomes a special case rather than a separate theory.

4.1 The Complex Exponential $e^{j\omega_0 n}$

In continuous time, $x(t) = e^{j\omega_0 t}$ transforms to a single impulse, $X(j\omega) = 2\pi\delta(\omega - \omega_0)$. In discrete time we might expect the same, but a single impulse cannot be correct: the DTFT must be periodic in ω with period 2π (Section 3.4). The transform must therefore contain an impulse not only at ω_0 but also at $\omega_0 \pm 2\pi$, $\omega_0 \pm 4\pi$, and so on:

$$e^{j\omega_0 n} \overset{\mathcal{F}}{\longleftrightarrow} X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} 2\pi\,\delta(\omega - \omega_0 - 2\pi l). \tag{23}$$

Verification by inverse transform. Substitute (23) into the synthesis equation (17) and integrate over one period:

$$x[n] = \frac{1}{2\pi}\int_{2\pi}\left[\sum_{l=-\infty}^{\infty} 2\pi\,\delta(\omega - \omega_0 - 2\pi l)\right] e^{j\omega n}\,d\omega.$$

Any interval of length 2π contains exactly *one* impulse of the train, say the one at $\omega = \omega_0 + 2\pi r$ for some integer r. The factors $1/2\pi$ and 2π cancel, and the sifting property of the impulse gives

$$x[n] = e^{j(\omega_0 + 2\pi r)n} = e^{j\omega_0 n}\underbrace{e^{j2\pi rn}}_{=1} = e^{j\omega_0 n},$$

as required. Note that the answer does not depend on which interval was chosen, precisely because $e^{j2\pi rn} = 1$ for integer n — the same fact that forced the impulse train in the first place.

4.2 A General Periodic Sequence

Let $x[n]$ be periodic with period N. By Section 2.5 it has a Fourier series

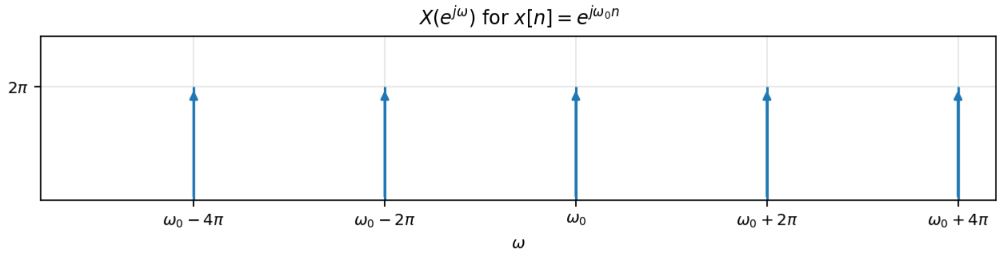


Figure 12: The DTFT of $e^{j\omega_0 n}$: impulses of area 2π at ω_0 and at every point 2π away from it.

$$x[n] = \sum_{k=\langle N\rangle} a_k e^{jk(2\pi/N)n}.$$

Each term is a complex exponential at frequency $k(2\pi/N)$, so applying (23) term by term and using linearity gives a train of impulses. Summing over all k from $-\infty$ to ∞ rather than over one period absorbs the $2\pi l$ replicas automatically, since a_k is itself periodic in k with period N. The result is

$$X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi a_k\, \delta\left(\omega - \frac{2\pi k}{N}\right). \tag{24}$$

Interpretation. The Fourier transform of a discrete-time periodic signal is an impulse train in ω, with impulses spaced $2\pi/N$ apart, and the area of the impulse at $\omega = 2\pi k/N$ is 2π times the kth Fourier series coefficient. So the DTFT of a periodic signal is constructed directly from its Fourier series coefficients, exactly as in continuous time. The two representations of Section 2 and Section 3 are the same object viewed differently.

Because $a_k = a_{k+N}$, the impulse pattern repeats every N impulses, that is, every 2π in ω — consistent with Section 3.4.

4.3 Worked Example 6: The Periodic Impulse Train

Problem. Find the DTFT of the discrete-time periodic impulse train

$$x[n] = \sum_{k=-\infty}^{\infty} \delta[n - kN].$$

Step 1: find the Fourier series coefficients. The signal is periodic with period N. Apply the analysis equation (18) over the interval $0 \le n \le N-1$, which contains exactly one impulse, at $n = 0$:

$$a_k = \frac{1}{N} \sum_{n=\langle N\rangle} x[n] e^{-jk(2\pi/N)n} = \frac{1}{N} \sum_{n=0}^{N-1} \delta[n] e^{-jk(2\pi/N)n} = \frac{1}{N}.$$

The sifting property selects the $n = 0$ term, where the exponential equals 1. So $a_k = 1/N$ for *every* k: all harmonics are present with equal weight.

Step 2: write the Fourier series. Substituting into the synthesis equation,

$$x[n] = \sum_{k=\langle N\rangle} \frac{1}{N} e^{jk(2\pi/N)n}.$$

Step 3: apply the transform of a periodic signal. Substituting $a_k = 1/N$ into (24),

$$\sum_{k=-\infty}^{\infty} \delta[n-kN] \overset{\mathcal{F}}{\longleftrightarrow} \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{N}\right). \tag{25}$$

Step 4: interpret. An impulse train in time transforms to an impulse train in frequency. The spacings are reciprocal: impulses every N samples in n produce impulses every $2\pi/N$ radians in ω. Making the time-domain train sparser makes the frequency-domain train denser, the same inverse width relationship seen in Examples 3 and 5.

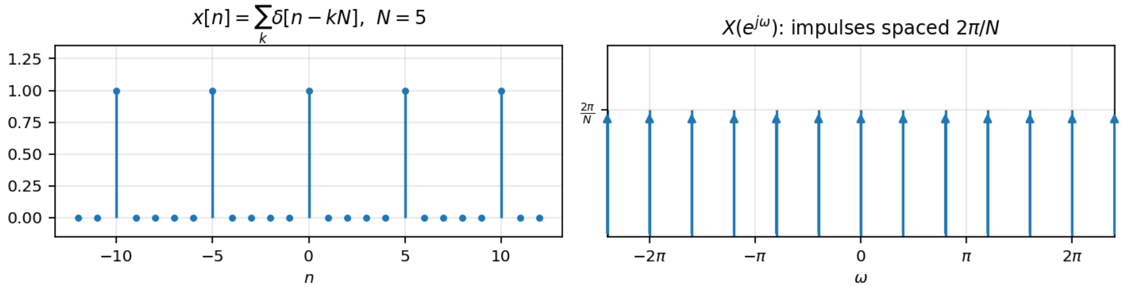


Figure 13: The periodic impulse train with $N = 5$ and its DTFT: impulses of area $2\pi/N$ spaced $2\pi/N$ apart in ω.

5 Properties of the Discrete-Time Fourier Transform

Throughout this section $x[n] \overset{\mathcal{F}}{\leftrightarrow} X(e^{j\omega})$ and $y[n] \overset{\mathcal{F}}{\leftrightarrow} Y(e^{j\omega})$.

5.1 Periodicity

$$X(e^{j\omega}) = X(e^{j(\omega+2\pi)}). \tag{26}$$

Proved in Section 3.4. This property has no continuous-time counterpart and is the source of most of the differences that follow.

5.2 Linearity

$$ax_1[n] + bx_2[n] \overset{\mathcal{F}}{\longleftrightarrow} aX_1(e^{j\omega}) + bX_2(e^{j\omega}). \tag{27}$$

This follows immediately from the linearity of the sum in (18).

5.3 Time and Frequency Shifting

$$x[n-n_0] \overset{\mathcal{F}}{\longleftrightarrow} e^{-j\omega n_0} X(e^{j\omega}), \qquad e^{j\omega_0 n} x[n] \overset{\mathcal{F}}{\longleftrightarrow} X(e^{j(\omega-\omega_0)}). \tag{28}$$

Proof of the time-shift property. Apply the analysis equation to $x[n-n_0]$ and substitute $m = n-n_0$:

$$\sum_{n=-\infty}^{\infty} x[n-n_0]e^{-j\omega n} = \sum_{m=-\infty}^{\infty} x[m]e^{-j\omega(m+n_0)} = e^{-j\omega n_0} \sum_{m=-\infty}^{\infty} x[m]e^{-j\omega m}.$$

A delay does not change the magnitude of the spectrum; it adds a linear term $-\omega n_0$ to the phase.

Worked Example 7: Turning a Lowpass Filter into a Highpass Filter

Problem. Let $H_{lp}(e^{j\omega})$ be the frequency response of a lowpass filter with cutoff frequency ω_c. Construct a highpass filter from it.

Step 1: identify where the high frequencies are. By Section 3.4, high frequencies in discrete time are concentrated near $\omega = \pi$ and other odd multiples of π, while the lowpass passband sits near $\omega = 0$ and other even multiples of π.

Step 2: shift by half a period. Shifting the response by π moves the passband from the even multiples of π to the odd ones. Using the frequency-shifting property with $\omega_0 = \pi$,

$$H_{hp}(e^{j\omega}) = H_{lp}(e^{j(\omega-\pi)}). \tag{29}$$

Step 3: identify the resulting cutoff. The original passband $|\omega| \leq \omega_c$ maps to the band centred on π extending from $\pi - \omega_c$ to $\pi + \omega_c$. So H_{hp} is a highpass filter with cutoff frequency $\pi - \omega_c$: a *narrow* lowpass filter becomes a *wide* highpass filter.

Step 4: the effect in the time domain. By (28) read in the other direction, multiplying a frequency response by $e^{j\pi n} = (-1)^n$ shifts it by π. So the highpass impulse response is obtained by alternating the signs of the lowpass impulse response: $h_{hp}[n] = (-1)^n h_{lp}[n]$.

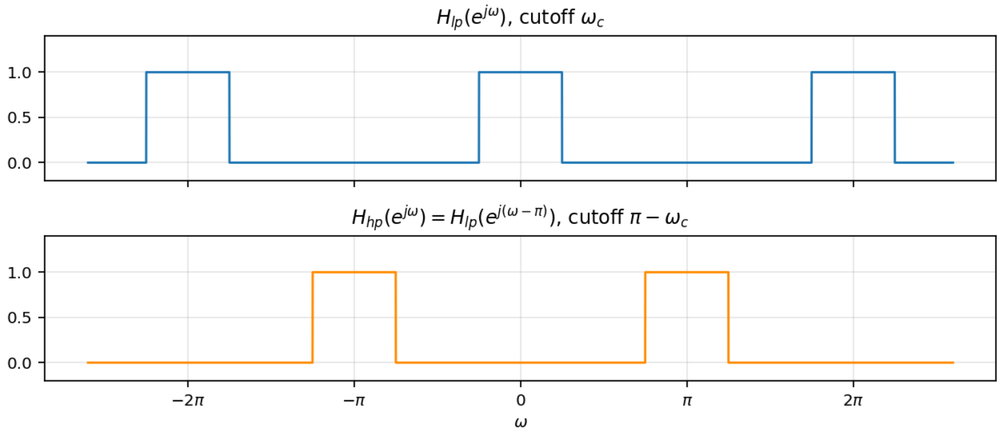


Figure 14: Shifting a lowpass frequency response by one half period (π) produces a highpass response with cutoff $\pi - \omega_c$.

5.4 Conjugation and Conjugate Symmetry

$$x^*[n] \overset{\mathcal{F}}{\longleftrightarrow} X^*(e^{-j\omega}), \qquad x[-n] \overset{\mathcal{F}}{\longleftrightarrow} X(e^{-j\omega}). \tag{30}$$

If $x[n]$ is **real**, then $X(e^{-j\omega}) = X^*(e^{j\omega})$, which splits into

$$\underbrace{\begin{cases} \mathrm{Re}\{X(e^{j\omega})\} \\ |X(e^{j\omega})| \end{cases}}_{\text{even in } \omega} \qquad \underbrace{\begin{cases} \mathrm{Im}\{X(e^{j\omega})\} \\ \sphericalangle X(e^{j\omega}) \end{cases}}_{\text{odd in } \omega}$$

Further, if $x[n]$ is real and even then $X(e^{j\omega})$ is real and even; if $x[n]$ is real and odd then $X(e^{j\omega})$ is purely imaginary and odd. The even and odd parts of a real $x[n]$ transform to the real and imaginary parts of $X(e^{j\omega})$ respectively:

$$\mathcal{E}v\{x[n]\} \overset{\mathcal{F}}{\longleftrightarrow} \mathrm{Re}\{X(e^{j\omega})\}, \qquad \mathcal{O}d\{x[n]\} \overset{\mathcal{F}}{\longleftrightarrow} j\,\mathrm{Im}\{X(e^{j\omega})\}.$$

These symmetries were visible in Example 4, where a real even pulse gave a real even transform.

5.5 Differencing and Accumulation

Differencing. Applying linearity and the time-shift property to $x[n] - x[n-1]$,

$$\mathcal{F}\{x[n] - x[n-1]\} = X(e^{j\omega}) - e^{-j\omega} X(e^{j\omega}),$$

so

$$x[n] - x[n-1] \xleftrightarrow{\mathcal{F}} \left(1 - e^{-j\omega}\right) X(e^{j\omega}). \tag{31}$$

Accumulation. The running sum $y[n] = \sum_{m=-\infty}^{n} x[m]$ is the inverse operation: it satisfies $y[n] - y[n-1] = x[n]$. Inverting (31) suggests dividing by $1 - e^{-j\omega}$, but that alone is incomplete, because differencing destroys any constant offset and the inverse must restore it. The correct result carries an additional impulse train accounting for the average (dc) value:

$$\sum_{m=-\infty}^{n} x[m] \xleftrightarrow{\mathcal{F}} \frac{1}{1 - e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{l=-\infty}^{\infty} \delta(\omega - 2\pi l). \tag{32}$$

Here $X(e^{j0}) = \sum_n x[n]$ is the sum of all the samples. If that sum is zero the impulse term vanishes and the accumulation is a well-behaved finite-energy signal.

5.6 Time Expansion

In continuous time, scaling gives $x(at) \leftrightarrow \frac{1}{|a|} X(j\omega/a)$. In discrete time $x[an]$ is not generally defined, because an need not be an integer. Two separate observations replace the continuous-time property:

- We *cannot* slow a signal down by choosing $a < 1$, since $x[n/a]$ is undefined at most n.
- Choosing an integer a other than ± 1, for instance $x[2n]$, does *not* speed the signal up in the continuous-time sense: because n takes only integer values, $x[2n]$ consists of the even-indexed samples of $x[n]$ alone, and the odd-indexed samples are lost.

The operation that does have a clean transform is expansion by inserting zeros. For a positive integer k, define

$$x_{(k)}[n] = \begin{cases} x[n/k], & n \text{ a multiple of } k, \\ 0, & n \text{ not a multiple of } k. \end{cases} \tag{33}$$

Derivation. Apply the analysis equation. Since $x_{(k)}[n] = 0$ unless $n = rk$ for some integer r, only those terms survive; substituting $n = rk$ and using $x_{(k)}[rk] = x[r]$,

$$X_{(k)}(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x_{(k)}[n] e^{-j\omega n} = \sum_{r=-\infty}^{\infty} x_{(k)}[rk] e^{-j\omega rk} = \sum_{r=-\infty}^{\infty} x[r] e^{-j(k\omega)r},$$

which is the analysis equation for $x[r]$ evaluated at frequency $k\omega$:

$$\boxed{x_{(k)}[n] \xleftrightarrow{\mathcal{F}} X(e^{jk\omega}).} \tag{34}$$

Interpretation. For $k > 1$ the signal is spread out and slowed down in time, while its transform is *compressed* in frequency. Since $X(e^{j\omega})$ is periodic with period 2π, $X(e^{jk\omega})$ is periodic with period $2\pi/k$. Figure 15 shows the case $k = 3$ applied to the rectangular pulse of Example 4.

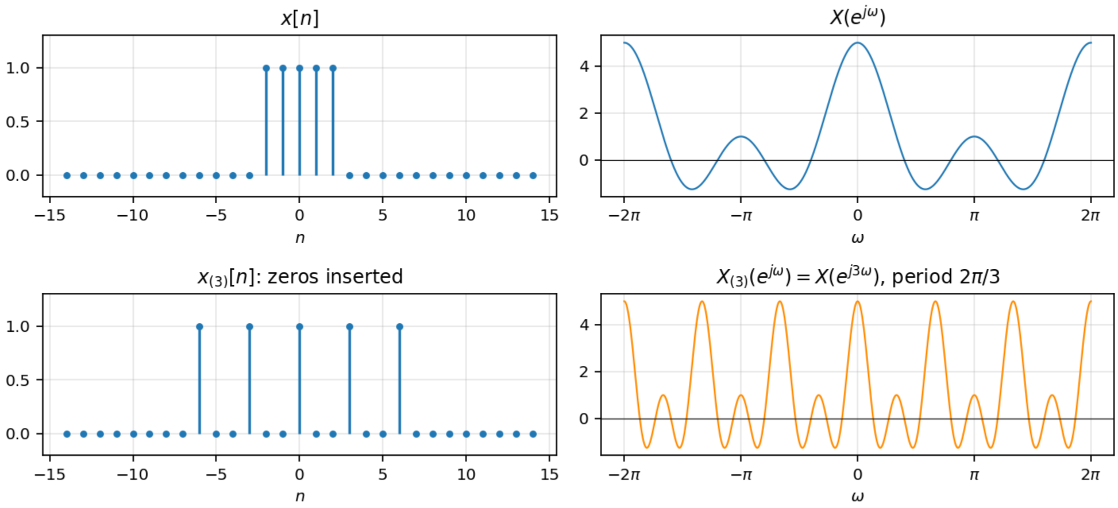


Figure 15: Time expansion by $k = 3$. Inserting two zeros between samples stretches the signal and compresses its transform, which now repeats with period $2\pi/3$ instead of 2π.

5.7 Differentiation in Frequency

Derivation. Differentiate both sides of the analysis equation (18) with respect to ω. Differentiating the summand gives a factor $-jn$:

$$\frac{d}{d\omega}X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n](-jn)e^{-j\omega n},$$

so $-jn\,x[n] \leftrightarrow \dfrac{d}{d\omega}X(e^{j\omega})$. Multiplying both sides by j and using $j \cdot (-j) = 1$:

$$n\,x[n] \xleftrightarrow{\mathcal{F}} j\,\frac{dX(e^{j\omega})}{d\omega}. \tag{35}$$

This property is used in Example 10 below to handle a repeated factor in a transform.

5.8 Parseval's Relation

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{2\pi} \left|X(e^{j\omega})\right|^2 d\omega. \tag{36}$$

The total energy of the sequence may be computed in either domain. The quantity $|X(e^{j\omega})|^2$ is the **energy-density spectrum**, describing how the energy is distributed over frequency.

5.9 The Convolution Property

Derivation. Let $y[n] = x[n] * h[n] = \sum_k x[k]h[n-k]$. Applying the analysis equation and exchanging the order of summation,

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty}\left[\sum_{k=-\infty}^{\infty} x[k]h[n-k]\right]e^{-j\omega n} = \sum_{k=-\infty}^{\infty} x[k]\left[\sum_{n=-\infty}^{\infty} h[n-k]e^{-j\omega n}\right].$$

The inner sum is the transform of a shifted $h[n]$, which by (28) equals $e^{-j\omega k}H(e^{j\omega})$. Pulling $H(e^{j\omega})$ out of the outer sum leaves the analysis equation for $x[k]$:

$$\boxed{x[n] * h[n] \xleftrightarrow{\mathcal{F}} X(e^{j\omega})H(e^{j\omega}).} \tag{37}$$

Convolution in time becomes multiplication in frequency. The function $H(e^{j\omega})$ is the **frequency response** of the LTI system with impulse response $h[n]$.

The dual property, for multiplication in time, is a *periodic* convolution over one period:

$$x[n]y[n] \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2\pi}\int_{2\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})\,d\theta. \tag{38}$$

Worked Example 8: A Pure Delay

Problem. Find the frequency response of the system with impulse response $h[n] = \delta[n-n_0]$, and the output for an arbitrary input.

Step 1: transform the impulse response. By the sifting property, the sum picks out $n = n_0$:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \delta[n-n_0]e^{-j\omega n} = e^{-j\omega n_0}.$$

Step 2: apply the convolution property. For any input $x[n]$ with transform $X(e^{j\omega})$,

$$Y(e^{j\omega}) = e^{-j\omega n_0}X(e^{j\omega}).$$

Step 3: interpret. $|H(e^{j\omega})| = 1$ at every frequency, so no frequency component is attenuated; only the phase changes, linearly with ω. Inverting recovers $y[n] = x[n-n_0]$, which is the time-shift property of Section 5.3 re-derived as a special case of convolution.

Worked Example 9: Convolution of Two Decaying Exponentials

Problem. An LTI system has impulse response $h[n] = \alpha^n u[n]$ with $|\alpha| < 1$. The input is $x[n] = \beta^n u[n]$ with $|\beta| < 1$ and $\alpha \neq \beta$. Find $y[n]$.

Step 1: transform both signals. From Example 3,

$$H(e^{j\omega}) = \frac{1}{1-\alpha e^{-j\omega}}, \qquad X(e^{j\omega}) = \frac{1}{1-\beta e^{-j\omega}}.$$

Step 2: multiply. By the convolution property,

$$Y(e^{j\omega}) = \frac{1}{(1-\alpha e^{-j\omega})(1-\beta e^{-j\omega})}.$$

Step 3: expand in partial fractions. Direct inversion of this product is awkward, but each factor alone is a transform we know. Seek

$$Y(e^{j\omega}) = \frac{A}{1-\alpha e^{-j\omega}} + \frac{B}{1-\beta e^{-j\omega}}. \tag{39}$$

To find A, multiply both sides by $(1-\alpha e^{-j\omega})$ and then set $e^{-j\omega} = 1/\alpha$, which makes the B term vanish:

$$\left.\frac{1}{1-\beta e^{-j\omega}}\right|_{e^{-j\omega}=1/\alpha} = \frac{1}{1-\beta/\alpha} = \frac{\alpha}{\alpha-\beta} = A.$$

Similarly, multiplying by $(1-\beta e^{-j\omega})$ and setting $e^{-j\omega} = 1/\beta$ eliminates A and gives

$$B = \frac{1}{1-\alpha/\beta} = \frac{\beta}{\beta-\alpha} = -\frac{\beta}{\alpha-\beta}.$$

Step 4: invert term by term. Each term in (39) has the form of Example 3, so by linearity

$$y[n] = \frac{\alpha}{\alpha-\beta}\alpha^n u[n] - \frac{\beta}{\alpha-\beta}\beta^n u[n] = \frac{1}{\alpha-\beta}\left(\alpha^{n+1} - \beta^{n+1}\right)u[n].$$

Step 5: sanity check. At $n = 0$ the expression gives $(\alpha - \beta)/(\alpha - \beta) = 1$, which matches direct convolution: $y[0] = x[0]h[0] = 1$.

Worked Example 10: The Repeated-Root Case $\alpha = \beta$

Problem. Repeat Example 9 when $\alpha = \beta$.

Step 1: see why the previous method fails. With $\alpha = \beta$ the partial fraction coefficients contain $\alpha - \beta = 0$ in the denominator. The transform is now a repeated factor:

$$Y(e^{j\omega}) = \left(\frac{1}{1 - \alpha e^{-j\omega}}\right)^2 = \left(1 - \alpha e^{-j\omega}\right)^{-2}.$$

Step 2: relate the squared factor to a derivative. Differentiate $\left(1 - \alpha e^{-j\omega}\right)^{-1}$ with respect to ω using the chain rule. Since $\frac{d}{d\omega}\left(1 - \alpha e^{-j\omega}\right) = j\alpha e^{-j\omega}$,

$$\frac{d}{d\omega}\left(1 - \alpha e^{-j\omega}\right)^{-1} = -\left(1 - \alpha e^{-j\omega}\right)^{-2}\left(j\alpha e^{-j\omega}\right).$$

Solving for the squared factor,

$$\left(1 - \alpha e^{-j\omega}\right)^{-2} = \frac{j}{\alpha} e^{j\omega} \frac{d}{d\omega}\left(1 - \alpha e^{-j\omega}\right)^{-1}. \tag{40}$$

Step 3: apply the differentiation property. From Example 3, $\alpha^n u[n] \leftrightarrow (1 - \alpha e^{-j\omega})^{-1}$. By (35),

$$n\alpha^n u[n] \overset{\mathcal{F}}{\longleftrightarrow} j\frac{d}{d\omega}\left(1 - \alpha e^{-j\omega}\right)^{-1}.$$

So the bracketed part of (40) is the transform of $n\alpha^n u[n]$, and

$$Y(e^{j\omega}) = \frac{1}{\alpha} e^{j\omega} \cdot \mathcal{F}\left\{n\alpha^n u[n]\right\}.$$

Step 4: apply the time-shift property. The factor $e^{j\omega}$ is a time *advance* by one sample, since $e^{-j\omega n_0}$ with $n_0 = -1$. Therefore

$$y[n] = \frac{1}{\alpha}(n+1)\alpha^{n+1} u[n+1] = (n+1)\alpha^n u[n+1].$$

Step 5: simplify the step function. The factor $(n+1)$ vanishes at $n = -1$, which is the only sample where $u[n+1]$ and $u[n]$ differ. The signal is therefore still zero prior to $n = 0$, and

$$\boxed{y[n] = (n+1)\alpha^n u[n].} \tag{41}$$

Step 6: cross-check against Example 9. Taking $\beta \to \alpha$ in the Example 9 answer gives the indeterminate form $0/0$; applying l'Hôpital's rule in β to $(\alpha^{n+1} - \beta^{n+1})/(\alpha - \beta)$ yields $(n+1)\alpha^n$, agreeing with (41).

5.10 Summary of Properties

6 Comparison of the Fourier Representations

6.1 The Four Representations Side by Side

6.2 Discrete in One Domain, Periodic in the Other

Reading Table 6 by the *character* of each domain rather than by its formula reveals a single organising principle.

The rule is:

Property	**Aperiodic signal**	**Fourier transform**
Periodicity	$x[n]$	$X(e^{j\omega}) = X(e^{j(\omega+2\pi)})$
Linearity	$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$
Time shifting	$x[n - n_0]$	$e^{-j\omega n_0} X(e^{j\omega})$
Frequency shifting	$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega-\omega_0)})$
Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$
Time reversal	$x[-n]$	$X(e^{-j\omega})$
Time expansion	$x_{(k)}[n]$	$X(e^{jk\omega})$
Convolution	$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$
Multiplication	$x[n]y[n]$	$\frac{1}{2\pi}\int_{2\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$
Differencing	$x[n] - x[n-1]$	$(1 - e^{-j\omega})X(e^{j\omega})$
Accumulation	$\sum_{k=-\infty}^{n} x[k]$	$\frac{1}{1-e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_l \delta(\omega - 2\pi l)$
Differentiation in frequency	$nx[n]$	$j\, dX(e^{j\omega})/d\omega$

Table 4: Properties of the discrete-time Fourier transform (Oppenheim, Willsky & Nawab, Table 5.1).

Signal	**Fourier transform**
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
$x[n] = 1$	$2\pi \sum_l \delta(\omega - 2\pi l)$
$e^{j\omega_0 n}$	$2\pi \sum_l \delta(\omega - \omega_0 - 2\pi l)$
$a^n u[n]$, $\lvert a\rvert < 1$	$\dfrac{1}{1 - ae^{-j\omega}}$
$(n+1)a^n u[n]$, $\lvert a\rvert < 1$	$\dfrac{1}{(1 - ae^{-j\omega})^2}$
$u[n]$	$\dfrac{1}{1 - e^{-j\omega}} + \pi \sum_l \delta(\omega - 2\pi l)$
$x[n] = 1$ for $\lvert n\rvert \le N_1$, else 0	$\dfrac{\sin[\omega(N_1 + \frac{1}{2})]}{\sin(\omega/2)}$
$\dfrac{\sin(Wn)}{\pi n}$, $0 < W < \pi$	1 for $0 \le \lvert\omega\rvert \le W$, 0 for $W < \lvert\omega\rvert \le \pi$
$\sum_k \delta[n - kN]$	$\dfrac{2\pi}{N} \sum_k \delta\left(\omega - \dfrac{2\pi k}{N}\right)$

Table 5: Basic discrete-time Fourier transform pairs (Oppenheim, Willsky & Nawab, Table 5.2). All frequency-domain expressions are periodic in ω with period 2π.

	Periodic signal (Fourier series)	**Aperiodic signal (Fourier transform)**
Continuous time	$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}$ $a_k = \dfrac{1}{T}\int_{\langle T\rangle} x(t)e^{-jk(2\pi/T)t}dt$	$x(t) = \dfrac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$ $X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$
Discrete time	$x[n] = \sum_{k=\langle N\rangle} a_k e^{jk(2\pi/N)n}$ $a_k = \dfrac{1}{N}\sum_{n=\langle N\rangle} x[n]e^{-jk(2\pi/N)n}$	$x[n] = \dfrac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$ $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$

Table 6: The four Fourier representations. The two entries in the bottom row are the subject of this lecture.

Representation	Time domain	Frequency domain
CT Fourier series	continuous, periodic	discrete, aperiodic
CT Fourier transform	continuous, aperiodic	continuous, aperiodic
DT Fourier series	discrete, periodic	discrete, periodic
DT Fourier transform	discrete, aperiodic	continuous, periodic

Table 7: Character of each domain. Periodicity in one domain forces discreteness in the other, and vice versa.

- **Periodic in one domain ⇒ discrete in the other.** A periodic $x(t)$ has a discrete spectrum a_k; a periodic $X(e^{j\omega})$ corresponds to a discrete $x[n]$.
- **Discrete in one domain ⇒ periodic in the other.** A discrete $x[n]$ has a spectrum periodic in ω; a discrete spectrum a_k corresponds to a periodic signal.

The discrete-time Fourier series is the only case where both domains are discrete *and* periodic, which is why both of its equations are finite sums. The continuous-time Fourier transform is the only case where neither domain is, which is why both of its equations are integrals over an infinite range.

6.3 What Is Specific to Discrete Time

The following points have no continuous-time counterpart and account for most of the differences developed in this lecture.

	Continuous time	Discrete time
Distinct harmonics $e^{jk\omega_0 \cdot}$	infinitely many	exactly N
Highest frequency	unbounded	$\omega = \pi$
Effect of increasing ω	faster oscillation, without limit	faster up to π, slower after
Fourier series convergence	conditions apply; Gibbs phenomenon	finite sum; always exact
Which transform equation is finite	analysis, over one period	synthesis, over 2π
Spectrum periodicity	none	always period 2π
Time scaling $x(at)$ or $x[an]$	defined for all real a	sample selection or zero insertion only

Table 8: Points where discrete-time Fourier analysis departs from the continuous-time case.

6.4 Threads Running Through the Lecture

Three ideas recurred, and it is worth collecting them.

Sampling a fixed envelope. In Example 1 the coefficients Na_k of a periodic square wave were samples of a fixed envelope, spaced $2\pi/N$ apart. Section 3.1 turned that observation into the derivation of the DTFT, and Example 4 confirmed it: the DTFT of a single rectangular pulse *is* that envelope.

Width in one domain against width in the other. A slowly decaying $a^n u[n]$ produced a narrow spectral peak (Example 3); a narrow lowpass band produced a widely spread sequence (Example 5); a sparse impulse train produced a dense one (Example 6); and time expansion by k compressed the spectrum by k (Section 5.6). These are all the same statement.

The role of π. Because $\omega = \pi$ is the highest discrete-time frequency, it is the natural centre for high-frequency behaviour: negative a in Example 3 put the spectral peak there, and shifting a lowpass filter by exactly π converted it to a highpass filter (Example 7).

7 Guide to the Next Lecture: The Discrete Fourier Transform

This section is a preview. It states where the next lecture goes and why, without deriving anything; the derivations belong to that lecture.

7.1 Why Neither Representation So Far Is Computable

Suppose we want a machine to compute a spectrum. Examine what each representation of this lecture would require.

The DTFT is not computable. The analysis equation (18) is an *infinite* sum over n, and its result $X(e^{j\omega})$ is a *continuous* function of ω. A computer can neither evaluate infinitely many terms nor store infinitely many values of ω. The synthesis equation (17) is worse still, being an integral.

The DTFS is computable, but only for periodic signals. By contrast, both equations of the DTFS pair, (8) and (9), are finite sums of exactly N terms, over N discrete values of n and N discrete values of k. This is precisely the situation identified in Table 7: the DTFS is the only representation in which *both* domains are discrete and finite. A machine can hold it exactly.

The catch is that the DTFS applies only to periodic signals, while real data — a recorded sound, a measured voltage, an image row — is a finite block of samples that is not periodic.

7.2 What the Discrete Fourier Transform Is

The discrete Fourier transform resolves this by taking the computable structure of the DTFS and applying it to a finite block of N samples $x[0], x[1], \ldots, x[N-1]$.

Start from the DTFT of the finite block, where the sum is now finite because the data is:

$$X(e^{j\omega}) = \sum_{n=0}^{N-1} x[n] e^{-j\omega n}.$$

This is still a continuous function of ω. The remaining step is to evaluate it at only finitely many frequencies, sampling uniformly across one period at $\omega_k = 2\pi k/N$. Writing $X[k]$ for $X(e^{j\omega_k})$ gives the pair

$$\boxed{X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)nk}, \qquad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(2\pi/N)nk},}$$

for $k, n = 0, 1, \ldots, N-1$. Both the signal and its spectrum are now finite lists of N numbers. Nothing is continuous and nothing is infinite, so the pair can be evaluated exactly by a machine — which is the whole reason the DFT, rather than the DTFS or the DTFT, is what software actually computes.

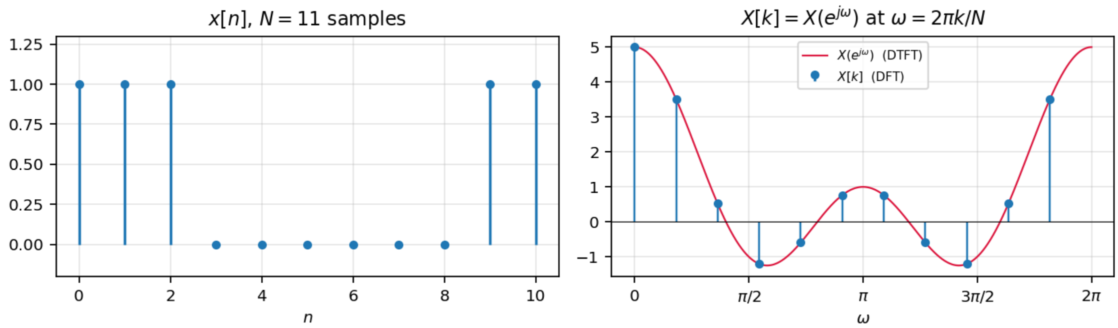


Figure 16: The DFT as samples of the DTFT. The continuous curve is $X(e^{j\omega})$ for the N-sample block on the left; the stems are the N values $X[k]$ retained by the DFT, taken at $\omega = 2\pi k/N$.

7.3 How It Connects to This Lecture

Two relationships tie the DFT to what has already been derived, and the next lecture establishes both.

The DFT samples the DTFT. By construction,

$$X[k] = X(e^{j\omega})\Big|_{\omega=2\pi k/N} = X\left(\frac{2\pi k}{N}\right).$$

This is the same envelope-and-samples relationship met twice already: in Example 1, where Na_k turned out to be samples of a fixed envelope spaced $2\pi/N$ apart, and in Section 3.1, where letting those samples merge produced the DTFT in the first place. The DFT simply travels that road in the opposite direction.

The DFT is N times the Fourier series coefficients. If $x[n]$ is treated as one period of a periodic sequence, then comparing the DFT with (9) gives

$$X[k] = Na_k.$$

Everything is implicitly periodic. Both relationships force a consequence worth anticipating: $X[k]$ is periodic in k with period N, and the $x[n]$ recovered by the inverse relation is periodic in n with period N. The DFT cannot represent a finite block *without* treating it as one period of something periodic. This is why the next lecture's shifting property involves a **circular shift** rather than an ordinary one, and why multiplying two DFTs corresponds to **periodic convolution** rather than the ordinary convolution of Section 5.9.

7.4 Why the DFT Is Practical: the FFT

Computed directly, the DFT costs on the order of N^2 operations. The fast Fourier transform is not a different transform but a divide-and-conquer algorithm for the same $X[k]$: it splits the sequence into even- and odd-indexed halves, transforms each, combines the results, and recurses. This reduces the cost to $O(N \log_2 N)$, which is what makes frequency-domain processing practical.

7.5 What to Carry Forward

- The orthogonality relation (6) reappears almost unchanged; it is what makes the inverse DFT work.
- The idea of a fixed envelope sampled at spacing $2\pi/N$ is the central picture.
- The properties of Section 5 mostly carry over, but shifts become circular and convolution becomes periodic, for the reason given above.

8 References

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  - Chapter 5
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